

Pushkal Mishra

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Biography

I am a researcher focused on developing scalable deep learning systems that perceive and reason about the physical world. My work spans multi-modal radar perception, autonomous driving, and sensor fusion, with an emphasis on building frameworks that unify simulation, real-world data, and large language models for robust scene understanding and decision-making. I am particularly interested in creating modular architectures that enable plug-and-play integration of diverse sensors while improving interpretability and generalization under challenging environmental conditions.

Education

University of California San Diego

PhD in Electrical and Computer Engineering

San Diego, CA, USA

Aug 2024 – Present

- Advisor: Prof. Dinesh Bharadia, [WCSNG Lab](#) 🔗
- Research Topics:
Radar Sensing, Multimodal Representation Alignment, Autonomous Driving

Indian Institute of Technology Hyderabad

B.Tech in Electrical Engineering, Minor in Computer Science

Hyderabad, India

Nov 2020 – May 2024

- PIs: Prof. Aditya Siripuram, Prof. Ayon Borthakur
- Graduated with CGPA: **9.33 / 10**

Royale Concorde International School

High School and Senior Secondary (CBSE)

Bangalore, India

Jun 2005 – Aug 2020

- Grade 12: **96.2%**, Grade 10: **90%**

Research Projects

Radar Foundation Model

Jan 2024 – Present

With Prof. Dinesh Bharadia at UCSD

- Built a radar-to-text foundation model for captioning, sim-to-real transfer, and interpretability, unifying radar perception and downstream tasks for autonomous driving.
- This framework enabled LLMs to reason over structured driving scene descriptions.
- Devised novel evaluation metrics tailored for generative radar models.
- Large-scale pre-training using Kubernetes, signal processing on radars, and multi-modal ML pipelines to improve generalization under noise and adverse conditions.
- **Skills:** Multi-modal ML pipelines, radar signal processing, large-scale pre-training, generative model evaluation.

Plug-n-Play Sensor Fusion Framework (P2SIF) for Autonomous Driving

Jun 2024 – Present

With Prof. Dinesh Bharadia at UCSD

- Developing a modular framework for fusing Radar, LiDAR, and Camera via text-based specialist models for LLM-driven decision making.
- Designing knowledge fusion mechanisms that decouple domain expertise (sensor processing) from generalist reasoning, to improve robustness and interpretability in autonomous driving.
- Currently evaluating system performance under adverse weather, plug-n-play sensor addition, and error injection cases, with focus on reliability and scalability.
- **Skills:** Modular sensor fusion, multi-sensor integration (Radar/LiDAR/Camera), LLM-driven reasoning, robustness and interpretability in autonomous driving.

Realistic Radar Simulation Framework for CARLA (C-Shenron)

Aug 2024 – Feb 2025

With Prof. Dinesh Bharadia at UCSD

- Developed a high-fidelity radar simulation framework integrated into CARLA simulator for autonomous driving.
- This simulator incorporated material-aware reflectivity and configurable radar parameters.
- Built scalable data collection pipelines (850K+ frames across diverse towns, weather, and sensor setups) using Kubernetes for large-scale parallel simulation.
- Trained and evaluated multi-radar fusion strategies along with ablations—demonstrating radar-camera fusion achieves performance comparable to LiDAR-camera baselines across CARLA leaderboard metrics.
- Published results in ACM SenSys'25 (Demo) and IEEE VTC'25 (Full Paper).

Graph Learning and Data Inpainting via Deep Neural Networks

Sept 2022 – Jan 2024

With Prof. Aditya Siripuram at IIT Hyderabad

- Devised an algorithm for simultaneous data restoration and constructing a pertinent graph structure.
- Implemented a closed-loop feedback mechanism for graph learning guided by the inpainting performance.
- Achieved highest F1-Score of 0.92 (learned and actual graph) among existing methods on synthetic data. Demonstrated robust performance under 10dB noise and above on temperature sensor network datasets.
- Published in IEEE Signal Processing Letters ([10.1109/LSP.2024.3501273](#)) and presented at ICASSP 2025.

New Training Algorithm for Deep Neural Networks

Jan 2023 – Jun 2024

With Prof. Ayon Borthakur at IIT Hyderabad

- Proposed a novel backpropagation-free training algorithm using a single forward pass with local loss functions.
- Eliminates gradient propagation and activation storage, yielding significant memory and computational savings.
- Implemented on Multilayer MLPs, CNNs and LSTMs while demonstrating competitive performance with backpropagation on benchmarks such as MNIST, CIFAR-10, CIFAR-100, and SVHN.
- Published in Transactions on Machine Learning Research (TMLR)

Selected Publications

A Realistic Radar Simulator for End-to-End Autonomous Driving in CARLA

Oct 2025

IEEE 102nd Vehicular Technology Conference

Chengdu, China

Satyam Srivastava*, Jerry Li*, **Pushkal Mishra***, Kshitiz Bansal, Dinesh Bharadia

Demo: Radar Simulator for CARLA

May 2025

23rd ACM Conference on Embedded Networked Sensor Systems

Irvine, USA

Pushkal Mishra, Satyam Srivastava, Jerry Li, Kshitiz Bansal, Dinesh Bharadia

Learning Using a Single Forward Pass

Jun 2025

Transactions on Machine Learning Research

Aditya Somasundaram*, **Pushkal Mishra***, Ayon Borthakur

Inpainting-Driven Graph Learning via Explainable Neural Networks

Apr 2025

IEEE Signal Processing Letters, Presented at IEEE ICASSP 2025

Hyderabad, India

Subbareddy Batreddy*, **Pushkal Mishra***, Yaswanth Kakarla, Aditya Siripuram

* Equal Contribution.

Awards

Electrical and Computer Engineering Department Fellowship

2024

Awarded full tuition support and stipend for five years of PhD at UC San Diego.

IEEE Signal Processing Cup

2023

Achieved rank 30 in the IEEE Signal Processing Cup challenge held worldwide.

Andy Grove Scholarship Award	2022
Recipient of the scholarship from Intel among 1800 competitive applicants to support undergraduate study.	
All India Rank of 1910 in JEE Advanced	2020
JEE Advanced examination among 1 Million candidates	
Merit Certificate in Physics (CBSE)	2020
Scored within top 0.1% nationwide among 1.7 Million candidates	

Skills

Programming Languages	Python, C/C++, MATLAB, Bash, Zsh, LaTeX, HTML, Verilog.
Tools & Frameworks	PyTorch, TensorFlow, Docker, Git, ROS2, MATLAB DSP/Deep Learning Toolbox, Cadence, LTSpice, NGSpice, KiCAD.
Systems Skills	Low-level systems programming, Scripting Automation, Kubernetes, Embedded Programming (ESP32, Jetson Nano), CARLA Simulation framework.

Work Experience

Signal Processing Intern	May 2023 – July 2023
<i>Texas Instruments</i>	<i>Bangalore</i>
- Designed and implemented a cross-talk cancellation algorithm for spatial audio playback. - Developed an efficient low-order filter approximation using bi-quad and all-pass filters within a 3 dB tolerance. - Proposed a new cross-talk cancellation architecture effective across different listener angles. - Introduced performance metrics for evaluating spatial image reconstruction and cross-talk cancellation efficiency. - Fully implemented solutions in MATLAB's Signal Processing and Deep Learning toolbox.	
Teaching Assistant	Aug 2023 – May 2024
<i>With Prof. Aditya Siripuram</i>	<i>IIT Hyderabad</i>
- Assistant for both Linear Systems & Signal Processing and Probability for AI. - Conducted tutorial sessions for students to solve assignment and exam doubts. - Formulated questions for bi-weekly quizzes and monthly exams.	
Senior Club Member	June 2021 – May 2023
<i>Epoch(ML) and Robotix Club</i>	<i>IIT Hyderabad</i>
- Built a mini-bipedal robot from scratch using 3D-printed parts, servo motors, and Arduino drivers. - Developed hand gesture recognition using motion sensors and DNN for time-series action classification. - Conducted workshops and hands-on sessions on Arduino and motor drivers. - Organized hackathons and knowledge-sharing sessions for the IIT Hyderabad community. - Participated in research paper presentation sessions.	
Class Representative	Aug 2022 – May 2023
<i>EE Department, Undergraduate 2020 Batch</i>	<i>IIT Hyderabad</i>
Elected representative for the undergraduate batch of 2020.	

Selected Course Projects

Autonomous Driving with Ultra-Wide Band (UWB) and Aruco Marker	Apr 2025 – Jun 2025
<i>ECE-148: Intro to Autonomous Driving Project at UCSD (Team of 4)</i>	
Designed and implemented an autonomous car-following system integrating real-time UWB-based triangulation, IMU fusion, and Aruco marker vision for robust localization and tracking even under non-line-of-sight conditions.	

- Built ROS2 perception with: OAK-D Lite for marker detection, UWB based anchor triangulation, and Aruco-based following with curved-path obstacle navigation.
- Project webpage: <https://github.com/UCSD-ECMAE-148/148-spring-2025-final-project-team-10/> 

IEEE 802.11 PHY/MAC Implementation

Nov 2024 – Jan 2025

Course Projects in ECE 257A and ECE 257B at UCSD

- Implemented core PHY components of IEEE 802.11 standard, including modulation/demodulation, OFDM channel access, channel estimation, and packet management (preamble detection, synchronization, and data recovery).
- Designed and tested end-to-end packet construction and decoding pipeline, covering STF/LTF generation, Tx-Rx frequency/time synchronization, and CSI-based channel equalization.
- Extended to advanced OFDM receiver implementation: packet detection via auto/cross-correlation, CFO/SFO estimation & correction, pilot-based phase error mitigation, and BER benchmarking under QPSK and 16QAM.
- Evaluated robustness across different synchronization and channel estimation strategies, analyzing trade-offs in detection accuracy and bit error rate.

Anchor-based Wi-Fi CSI Localization

Aug 2024 – Dec 2024

ECE-257A: Modern Communication Networks Project at UCSD (Team of 3)

- Proposed a novel Anchor-based CNN architecture, incorporating anchor-assisted features to achieve sub-meter accuracy in dynamic multipath environments.
- Built an end-to-end data collection and preprocessing pipeline using Turtlebot, Raspberry Pi anchors, and SLAM for ground-truth alignment; performed CSI phase/magnitude calibration and ToF correction.
- Collected and trained on vast amounts of real-scenario data in dynamic environments.
- Demonstrated that DL with CSI significantly outperforms RSSI and classical approaches for indoor localization.